

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

**Pharm***Achieve*

# CANNABIS

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

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**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

**CB** Cannabinoid Receptor

**CBD** Cannabidiol

**CHS** Cannabinoid Hyperemesis Syndrome

**CINV** Chemotherapy-Induced Nausea and Vomiting

**CNS** Central Nervous System

**COPD** Chronic Obstructive Pulmonary Disease

**CYP** Cytochrome P450

**D/C** Discontinued

**DIN** Drug Identification Number

**DS** Dravet Syndrome

**HR** Heart Rate

**LGS** Lennox-Gastaut Syndrome

**NHP** Natural Health Product

**NT** Neurotransmitters

**PTSD** Post-traumatic Stress Disorder

**PD** Pharmacodynamics

**PK** Pharmacokinetics

**SCI** Spinal Cord Injury

**THC**  $\Delta^9$ -Tetrahydrocannabinol

**TSC** Tuberous Sclerosis Complex

**UGT** Uridine 5'-Diphospho-Glucuronosyltransferase

# AN INTRODUCTION TO CANNABIS

- Genus of flowering plants with **various species** (*Cannabis sativa*, *indica*, *ruderalis*), **routes of administration, formulations and effects** when used
  - Marijuana is a psychoactive drug made from the cannabis plant

Cannabis exerts its effects through compounds called **cannabinoids**, which **interact with the endocannabinoid system via cannabinoid receptors**

- Most common cannabinoids:
  - $\Delta$ -9-tetrahydrocannabinol (THC): partial agonist to CB1 and CB2
  - Cannabidiol (CBD): indirect CB activity
- Different cannabis plants have different ratios of CBD and THC

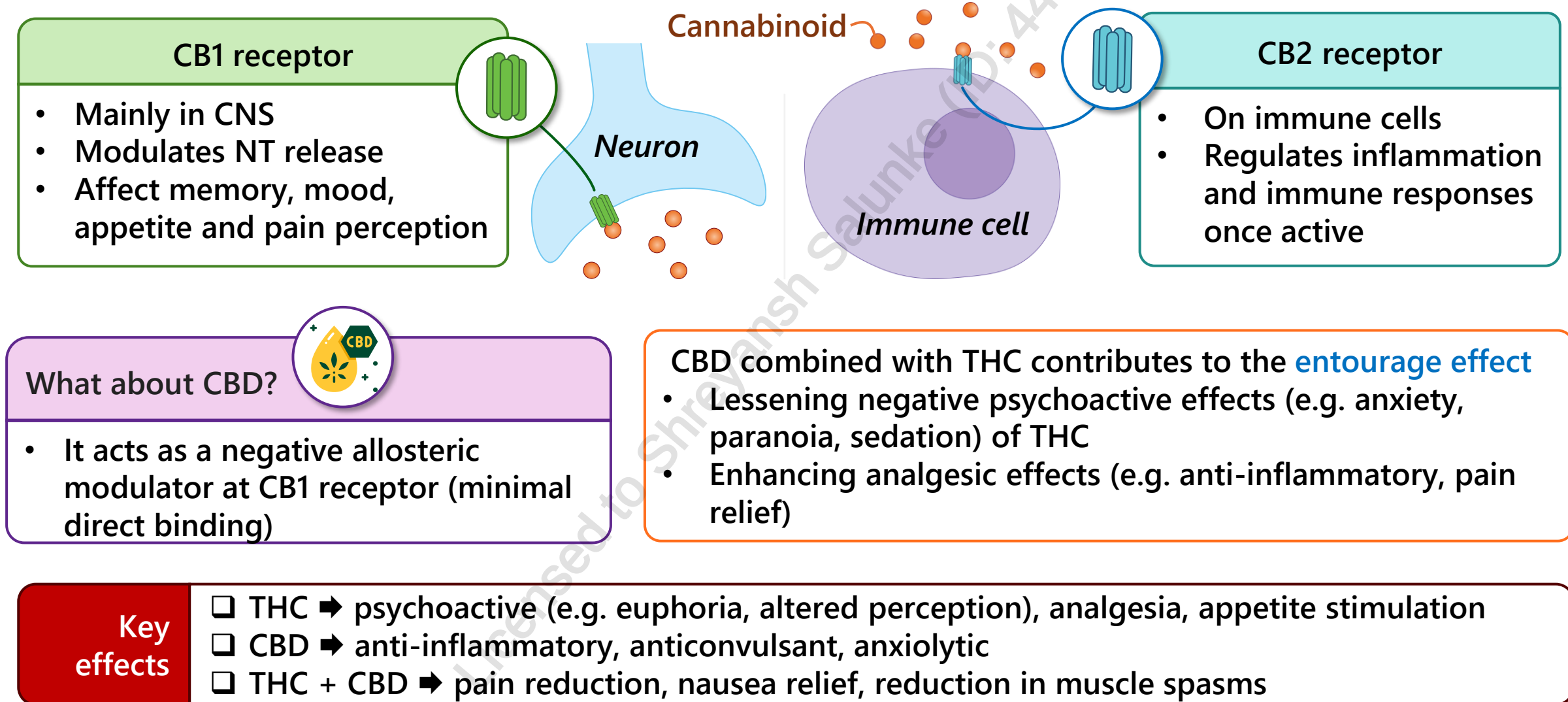


Sativa



Indica

# HOW DOES CANNABIS AFFECT THE BODY AND MIND?



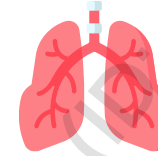
# SPECTRUM OF CANNABIS EFFECTS

## Neurological



- **Acute:** sedation, cognitive impairment, mood enhancement,
- **Long-term:** dependency, memory impairment, tolerance

## Respiratory



- **Acute:** smoking cannabis is cytotoxic
- **Long-term:** chronic bronchitis, increased airway resistance

## Musculoskeletal



- **Acute:** analgesic effects in acute injuries
- **Long-term:** unknown

## Psychiatric



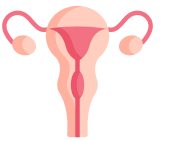
- **Acute:** panic attacks, potential psychosis
- **Long-term:** cannabis use disorder, early schizophrenia in patients at risk

## Gastrointestinal



- **Acute:** appetite stimulation, dry mouth, antiemetic effects
- **Long-term:** unknown

## Reproductive



- **Acute:** decreased sperm motility
- **Long-term:** irregular menstrual cycles, fetal growth restriction

## Cardiovascular



- **Acute:** increased HR, blood pressure changes
- **Long-term:** tachyarrhythmias

## Endocrine



- **Acute:** metabolic hormone modulation (e.g. increases ghrelin)
- **Long-term:** altered glycemic control in diabetic patients

## Ocular



- **Acute:** decreased intraocular pressure, dry eyes, red eyes
- **Long-term:** unknown

# CANNABINOID HYPEREMESIS SYNDROME (CHS)

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- Severe nausea, cyclic vomiting and abdominal pain secondary to chronic cannabis use
  - Pathophysiology is unknown
- Complications: weight loss, dehydration, electrolyte imbalances, heart rhythm abnormalities, seizures, kidney damage, mortality
- **Treatment: Complete cannabis cessation** is the most effective long-term treatment for CHS
  - **Symptom improvement:**
    - Rehydration - Intravenous fluid resuscitation
    - Improve CHS symptoms - Hot showers/ baths
    - Abdominal pain – acetaminophen or ibuprofen, avoid opioids
- Monitoring: electrolytes and kidney function

# RESTRICTIONS TO CANNABIS USE

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Recommend AGAINST USE of medical cannabinoids for most conditions due to lack of evidence of benefit and known harms (Strong)\*

\*Exceptions to the recommendation are on slide 11

## Not recommended/ Contraindications in patients with:

- ☐ Special populations: pregnancy/ planning pregnancy, breastfeeding, elderly, youth
- ☐ Psychiatric disorders: anxiety, PTSD, insomnia, psychotic disorders, schizophrenia, bipolar disorder, personal or family history of psychosis
- ☐ Cardiovascular disease (past event or risk factors)
- ☐ Respiratory comorbidities (use of dried and smoked cannabis)
- ☐ Personal or family history of substance use disorder
- ☐ Concurrent use of cannabis with other substances (e.g alcohol, tobacco)
- ☐ Cannabis use and driving

# ROUTES OF ADMINISTRATION AND THC COMPARISON

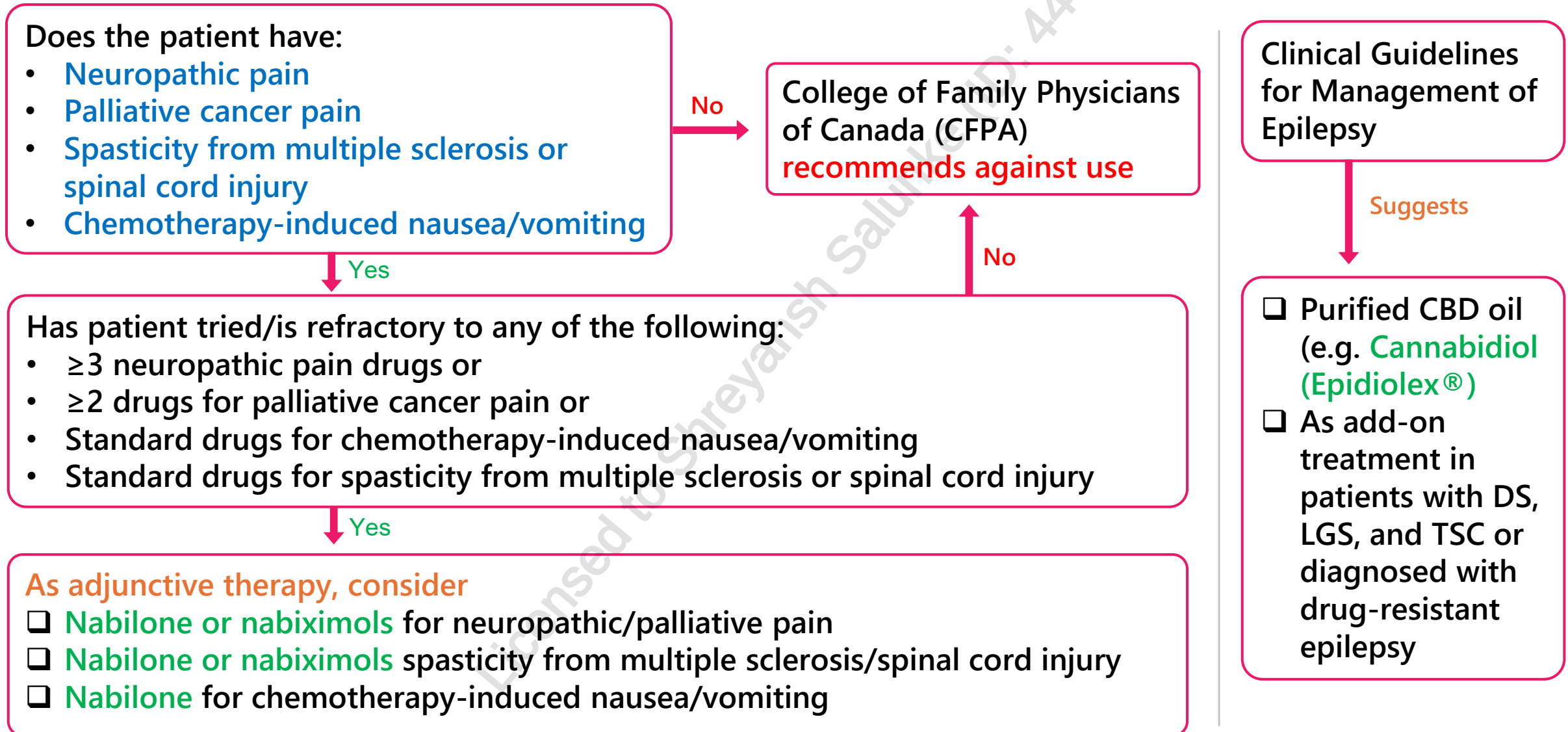
| Features                         | Smoking                                                                                                        | Vaporized                                                                                                                                                                    | Oral consumption                                                                                                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| THC Bioavailability              | 2- 56%                                                                                                         |                                                                                                                                                                              | 4-12%                                                                                                                                                                              |
| Onset                            | Almost immediately up to 30 minutes                                                                            |                                                                                                                                                                              | Delayed, 0.5-4 hours                                                                                                                                                               |
| Peak plasma concentration        | 6-10 minutes post-inhalation                                                                                   |                                                                                                                                                                              | 2-6 hours                                                                                                                                                                          |
| Duration of psychoactive effects | 2-4 hours*                                                                                                     | 2-3 hours*                                                                                                                                                                   | 12-24 hours                                                                                                                                                                        |
| Other comments                   | <ul style="list-style-type: none"> <li>Combustion is carcinogenic and causes a loss of cannabinoids</li> </ul> | <ul style="list-style-type: none"> <li>Highest extraction efficiency</li> <li>Absent combustion</li> <li>Maximizes cannabinoids</li> <li>No first pass metabolism</li> </ul> | <ul style="list-style-type: none"> <li>Decreased bioavailability due to first-pass hepatic metabolism</li> <li>Erratic/ unpredictable absorption</li> <li>Overdose risk</li> </ul> |

\* Some residual effects can last longer (up to 24 hours)

# RECOMMENDATIONS FOR NEW NON-MEDICAL CANNABIS USERS

| Disclaimer                                                                                                                                                                                                                                       | Inhaled use                                                                                                                                                                                                                                                                                                                                    | Oral consumption                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• PK: start low and go slow</li><li>• PD: variability in CB receptors, cannabinoid metabolism, prior exposure (tolerance)</li><li>• Drug factors: plant strain, potency, route of administration</li></ul> | <ul style="list-style-type: none"><li>• To minimize risk, start with low doses and stop if there are any significant or unwanted side effects</li><li>• Health Canada advises waiting <b>at least 10-20 minutes</b> between inhalations</li><li>• Products with over 20% THC (200 mg/g) increase the risk of serious adverse effects</li></ul> | <ul style="list-style-type: none"><li>• To reduce risk, start with low doses and stop if there are any significant or unwanted side effects</li><li>• Health Canada advises waiting <b>at least 30 minutes (preferably 3 hours)</b> between bites</li><li>• Use extra caution with cannabis products containing <math>\geq 10</math> mg of THC</li></ul> |

# GUIDELINE-BASED ALGORITHM FOR MEDICAL CANNABINOID USE



# MEDICAL CANNABINOIDS COMPARISON

|                                       | Nabilone (Cesamet®)                                                                                                                                               | Nabiximols (Sativex®)                                                                                                                                                                                                                         | Cannabidiol (Epidiolex®)                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Active Ingredients                    | <ul style="list-style-type: none"> <li>Synthetic THC</li> </ul>                                                                                                   | <ul style="list-style-type: none"> <li>1:1 of THC &amp; CBD</li> </ul>                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Purified CBD</li> </ul>                                                                                                                                                                                                                                                                                       |
| Formulation                           | <ul style="list-style-type: none"> <li>Oral capsules</li> </ul>                                                                                                   | <ul style="list-style-type: none"> <li>Buccal spray</li> </ul>                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Oral solution</li> </ul>                                                                                                                                                                                                                                                                                      |
| Cytochrome and drug-drug interactions | <ul style="list-style-type: none"> <li>Not metabolized by CYP</li> <li>Additive CNS depressant effects when taken with sedative medications or alcohol</li> </ul> | <ul style="list-style-type: none"> <li>Metabolized by CYP3A4, 1A2, 2C9, 2D6, 2C19</li> <li>Possible interaction with strong 3A4 and/ or 2C19 inducers/ inhibitors</li> <li>May reduce the effectiveness of hormonal contraceptives</li> </ul> | <ul style="list-style-type: none"> <li>Substrate of CYP3A4 and 2C19, decreased concentration when co-administered with strong inducer</li> <li>Potential to inhibit UGT1A9, UGT2B7, CYP1A2, CYP2C8, CYP2C9, and CYP2C19 at concentrations relevant to clinical use.</li> <li>Avoid use with alcohol due to the increased risk of sedation</li> </ul> |

## Non-medical cannabis

- Smoked cannabis induces CYP1A2
- Oral cannabis inhibits CYP2C19/3A4, space opioids by 4-6 hours

# MONITORING AND FOLLOW-UP

| Parameters             | Target                                                                 | Timeline                                                                                                                                           |
|------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Clinical response      | Improvements to pain, decreased spasticity and/or CINV                 | Follow-up should be done within 1-2 weeks of initiating medical cannabinoids to assess the safety, tolerability, and early improvement in symptoms |
| CNS and Cardiovascular | Monitor for tachycardia, BP fluctuations and respiratory complications |                                                                                                                                                    |
| Side effects           | Monitor for psychiatric symptoms, CNS depression and drug side effects |                                                                                                                                                    |

# LEGAL CANNABIS USE

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- $\geq 18$  years old is the federal minimum age to engage in cannabis activities
- **Note:** Federal legislation allows provincial/ territorial restrictions to set higher age requirements

## The Cannabis Act allows:

- Purchase dried or fresh cannabis and cannabis oil from a provincially licensed retailer, or federally licensed producers online in provinces and territories without a regulated retail system
- **Possess up to 30 g of legal cannabis**, dried or equivalent in non-dried form, in public
- **Share up to 30 g of legal cannabis** with other adults
- Grow, from licensed seed or seedlings, **up to 4 cannabis plants** per residence for personal use
- Permission to make cannabis products like food and drinks at home, if organic solvents are not being used to create concentrates
- 1 gram of dried cannabis is equivalent to: 5 grams of fresh cannabis, 15 grams of edibles, 70 grams of liquid product, 0.25 grams of concentrates (solid or liquid), or 1 cannabis plant seed

# CANADA'S LOWER-RISK CANNABIS USE GUIDELINES



Canada's Lower-Risk Cannabis Use Guidelines (LRCUG) are a groundbreaking, evidence-based public health initiative aimed at helping individuals minimize the health risks linked to cannabis use following its legalization in 2018.

1. The most effective way to avoid the risks of cannabis use is to abstain from use
2. Delaying cannabis use, at least until after adolescence, will reduce the likelihood or severity of adverse health outcomes regarding brain development
3. Use products with low THC content and high CBD:THC ratios
4. Synthetic (illegal) cannabis products, such as K2 and Spice, should be avoided
5. Avoid smoking burnt cannabis and choose safer methods, including vaporizers, e-cigarette devices, and edibles
6. If cannabis is smoked, avoid harmful practices such as inhaling deeply or breath-holding
7. Avoid frequent or intensive use, and limit consumption to occasional use
8. Do not drive or operate other machinery for at least 6 hours after using cannabis. Combining alcohol and cannabis should be avoided since it increases risk of impairment
9. People with a personal or family history of psychosis or substance use disorders, as well as pregnant women, should not use cannabis at all
10. Avoid combining any of the risk factors related to cannabis use. Multiple high-risk behaviours will amplify the likelihood or severity of adverse outcomes

# TAKEAWAY

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- Cannabis plant contains cannabinoids such as CBD and THC
- THC is associated with feelings of being 'high' due to its psychoactive nature
- CBD has anti-convulsant, anti-inflammatory and anti-anxiety effects
  - CBD does not interact with CB1 or CB2 receptors
- CHS: abdominal complications with nausea and profuse cyclic vomiting
  - Treatment: Cannabis cessation
  - Supportive care: intravenous fluids for dehydration, pain management (e.g. acetaminophen or ibuprofen), hot showers/ baths to improve CHS symptoms
- **Medical Cannabis Canadian Guidelines recommend against** prescribing medical marijuana (particularly smoked) as first-line cannabinoid due to biased studies and known harmful effects
  - If medical marijuana is to be used, nabilone or nabiximols are preferred
  - Nabiximols are metabolized by the CYP enzymes but nabilone is not
- There are currently **no pharmacological treatments for cannabis-use disorder**

# CASE 1

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KJ is a 35-year-old female patient with epilepsy who is stabilized on clobazam 20mg per day monotherapy. She works as a construction site manager, often working long hours on the job site, enforcing workplace safety, accepting deliveries and occasionally filling in when short-staffed to help her team meet deadlines. With a workplace history of heavy lifting, physically demanding shifts and using vibrating tools, she often smokes cannabis to help with her chronic lower back pain and general aches. Recently, after using a jackhammer to break up the pavement, she noticed sharp, burning pain accompanied by numbness and tingling in her hands, which prevented her from performing tasks that require fine motor movements. KJ arrives at the pharmacy today looking to get some information on medical cannabis, as her supervisor suggested it is safer than smoking and has helped him with similar symptoms. KJ consumes a case of beer a week, smokes 1-2 joints every evening, and does not exercise. She is sexually active and adherent to her hormonal contraceptive. KJ occasionally uses naproxen 500 mg twice daily and diclofenac extra-strength topical gel as needed, applied twice daily, for her lower back pain. KJ has no known allergies.



# CASE-BASED QUESTIONS

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1. Which of the following cannabis-derived pharmaceutical products are currently approved by Health Canada for specific medical indications?
  - a) Nabilone and Dronabinol
  - b) Nabilone and hemp
  - c) Nabiximols and kratom
  - d) Nabilone and Nabiximols
  
2. Which of the following cannabis products would be the SAFEST to try first?
  - a) A product from dispensary A with a 7:1 THC:CBD ratio
  - b) A product from dispensary B with a 5:2 THC:CBD ratio
  - c) A product from dispensary C with a 11:1 CBD:THC ratio
  - d) A product from dispensary D with 100mg of THC and no CBD

# CASE-BASED QUESTIONS

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3. Which of the following conditions is the most appropriate indication nabiximol in Canada?
- a) Fibromyalgia-related sleep disturbances
  - b) Spasticity in multiple sclerosis
  - c) Pediatric epilepsy
  - d) Rheumatoid arthritis pain
4. KJ's physician calls into the pharmacy, stating that the patient wants a formulation that she does not have to swallow. Which of the following would be the best recommendation?
- a) Nabilone
  - b) Nabiximols
  - c) Cannabidiol
  - d) Dronabinol

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# CHANGE LOG

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## July 2025

- Significant changes made throughout
- Information consolidated

## October 2025

- Content reviewed, no changes

## November 2025

- Slide 3: added CHS
- Slide 7: new slide on cannabis hyperemesis syndrome
- Slide 9: Added minor footnote
- Slide 10: Updated recommended waiting intervals between doses
- Slide 16: Updated takeaway to include CHS

## March 2026

- Slide 2: Updated NAPRA competencies
- Slide 10: Minor update
- Minor formatting changes throughout